ABSTRACT

Technology has advanced exponentially in last few decades, we have made machines so small we have made chips with size of individual transistors of size less than 10 nm that powers our daily life activities, many microscopic electromechanical devices also exist that work flawlessly.

For years we dreamt about nano scale robots that can move as per our command and execute various tasks such as moving and assembling into 3d structures.

In this paper I present a similar concept idea with designs and calculations that aligns with **modular robotics, programmable matter** and **nanotechnology** and more particularly to **self-assembling micro-scale or nano-scale robotic units** capable of forming reconfigurable three-dimensional structures using **electromagnetic interactions.**

Tesselation: arrangement of matter or polygons without gaps or space.

VISION

The motivation behind this project was to develop a microscopic bots that can move on their own, bond with other bots, share data and energy to each other, this can particularly be useful for

DESIGN LOG

The first design I designed was basically a elongated diamond shape with smooth edges looking more like a cylinder, later that design was discarded because, that shape allowed only two joints at max, it was difficult to organise that structure in 3d to form 3d structures, another reason that design was discarded was because it was difficult to make that structure in microscopic level as it was unsymmetrical

To solve all these issues I needed a symmetric structure that can attract and repel other particles in microscopic level and can pack more number of particles in a given space, and meeting the need of maximum number of joint for simple arrangement and dearrangement algorithm.

Simple to manufacture, symmetric about all 3 axes and maximum number of joints and design that was simpler to assemble in 3d to form structures.

A cube seems to be the perfect base structure that meets all above mentioned criterion,

Next challenge was to tackle the huge resistance of the coil, and I did it by increasing the cross-sectional area of the coil.

DESIGN

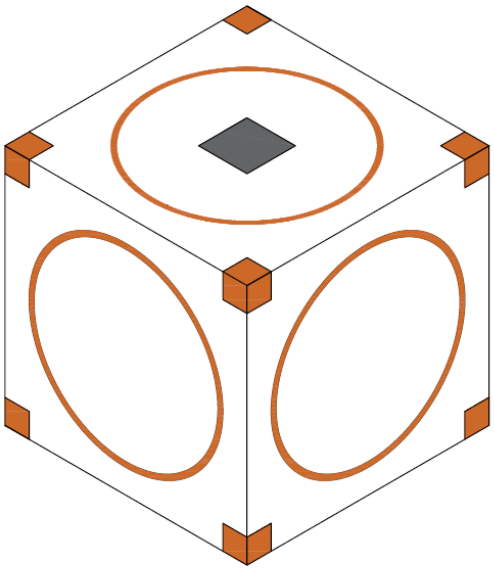

FIG 1

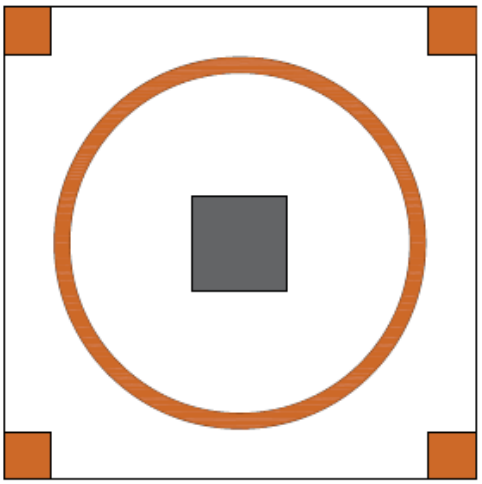

Fig 2

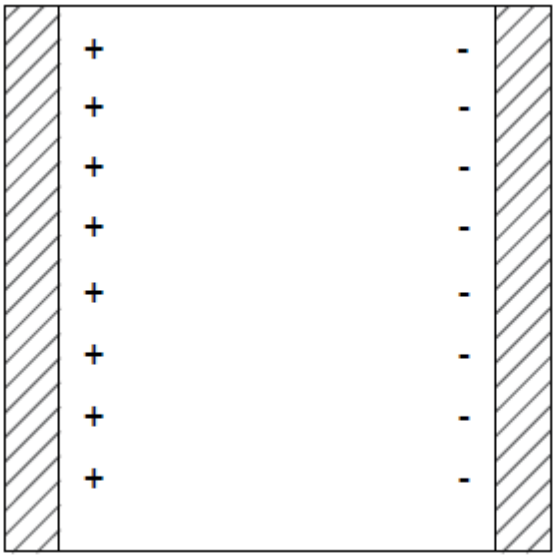

Fig 3

COILS

The bot has copper coils on each face. These coils serve two purpose,

1. **Magnetic Bond**: Attract and repel other bot or create magnetic bond, When current is passed through the coils, a magnetic field is generated, two cubes with opposite polarity, when close attract each other and create a bond. The coils can also be used to create push or pull force between bots, allowing them to move. The current in those coils can be made adjustable such that we can control the push/pull force or bond strength.

COMMUNICATION PADS

4 conductive pads can be placed on every edge of one face, these pads will be responsible for information transfer between bots, these pads serve two purpose,

1. Data transfer between bots: these pads will share crucial commands and data to specific bots ultimately establishing communication among bots which will be crucial for organised operations of the bots, alternatively many bots can come together, connect to each other and form a CPU with combined processing power of each bot.

2. Energy transfer between bots: these pads will be useful for transferring energy to bots that require more energy. For example, if bots form a spherical shape, the bots present at the centre of the sphere will need way more energy as compared to the bots present on outside region to hold the sphere together.

INTERNAL CAPACITOR

The Bot may include **conductive plates positioned on opposing faces**, forming an internal capacitor which will be used for **temporarily storing harvested energy**.

This capacitive storage will be useful when,

1. Multiple bots are connected simultaneously.
2. Higher instantaneous power is required.

The dielectric or internal material between the plates may vary, allowing **material-based customization**, including implementations using PCB-etched conductors or metallic substrates.

INTERNAL CONTROL (BRAIN)

Each Bot may include a **miniaturized CPU, microcontroller, or logic circuitry**, directly fabricated or etched onto the Bot.

The internal controller enables:

- Local decision-making
- Coil activation timing
- Communication with neighbouring Bots or external controller.

Self-Assembly and Reconfiguration

A plurality of Bots autonomously assemble into predefined or dynamically changing 3D structures based on:

- Coil activation sequences
- External magnetic field parameters
- Local interactions between Bots

Structures may be **reconfigured in real time** by modifying the external field or activation logic.

powering the bots

An external oscillating magnetic field will induce current in the coils of these bots. The same coils that are used for bonding will be used to power the bots, suppose at time t = 0 second the bot will use the coils for harvesting electrical energy and that energy will be stored in its internal capacitor, at t = 1 second the stored energy will be used to operate the coils for bonding purposes, the cycle of harvesting and using the harvested energy in sync will be used by the bots.

If certain bots are in critical positions and its impossible for them to stop magnetic bond to harvest magnetic field, in such cases sourrounding bots will share energy to that particular bot using the communication pads.

Current Limitations

1. The capacitor cannot store enough energy required by the bot.
2. Magnetic force between bots is poor.
3. Energy generated by the coils is minute and not enough for powering the bot.
4. Without proper arranging algorithm it will be a complete mess.

Some Preliminary Calculations

1. Calculation of magnetic for between two bots at certain distance
2. Current required for generating that magnetic force
3. Current induced in the coils via external magnetic field
4. Heating caused by these currents
5. Energy storable in the capacitor.

Similar Projects

1. M-Blocks
2. Micro solarcell based robot by Upenn and U-M